

Task 1:

```
karthik@warthikeya:~$ nano command_executor.c
karthik@warthikeya:~$ gcc command_executor.c -o command_executor
karthik@warthikeya:~$ ./command_executor
Enter a Linux command: ls

Parent Process
Parent PID : 1316
Child PID : 1319

Child Process
Child PID : 1319
Parent PID : 1316
Executing command: ls
OSSP command_executor command_executor.c fork_example.c process process.c process1.c

Child process completed.
karthik@warthikeya:~$ uname -a
```

Task 2: Investigation of Hardware Resources and Operating System Services

Introduction

In this experiment, Linux terminal commands were used to study different hardware resources managed by the operating system. Commands such as `uname -a`, `lscpu`, `ps`, and `top` were executed to view system information, processor details, running processes, and system performance. This experiment helped in understanding how the operating system provides a simple way to interact with the computer hardware.

Command: `uname -a`

Output:

```
karthik@warthikeya:~$ uname -a
Linux warthikeya 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
```

Explanation:

The `uname -a` command displays information about the operating system. From the output, it is observed that the system is running Ubuntu Linux on WSL2 with a 64-bit architecture. It also displays the kernel version and system information, which helps identify the operating system environment.

Command: `lscpu`

Output:

```

# cat /sys/devices/virtual/dmidecode/meminfo
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 39 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 24
On-line CPU(s) List: 0-23
Vendor ID: GenuineIntel
Model name: 13th Gen Intel(R) Core(TM) i7-13700HX
CPU family: 6
Model: 191
Thread(s) per core: 2
Core(s) per socket: 12
Stepping: 1
BogoMIPS: 46097.99
Flags: fpu_vme de pae tsc mtr par mce cml apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse2 ss ht syscall nx pdpeiblt rdtscp lm constant_tsc rep_good nopl topology tsc_reliable nonstop_tsc cpuid tsc_known_freq mmi1 pni pclmulqdq
vmx_ssse3d cx16 clflush_opt sse4_1 sse4_2 movbe popcnt tsc_deadline_timer avx avic rdt_intel hypervisor_lvl_1 sha_ni smonoprocessor ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow split_mpid smpl_mfghash tsc_adjust mwaitx v2
2 smp bnl2 errata indexed pdsed asm clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 avx5vnni vmw umip w
altpgm gdt vms vpclmulqdq rdpid movdiri movdir64b fpmv md_clear serialize_l3 flush_l3d arch_capabilities

Virtualization features:
Virtification: VT-x
Hypervisor vendor: Microsoft
Virtualization type: full
Caches (sum of all):
L1d: 576 KIB (12 instances)
L1i: 384 KIB (12 instances)
L2: 15 MIB (12 instances)
L3: 30 MIB (1 instance)
NUMA:
NUMA node(s): 1
NUMA node CPU(s): 0-23
Vulnerabilities:
Gather data sampling: Not affected
Ghostwrite: Not affected
Indirect target selection: Not affected
Itlb multihit: Not affected
L1tf: Not affected
Meltdown: Not affected
Mmio stale data: Not affected
Old microcode: Not affected
Reg file data sampling: Mitigation; Clear Register File
Retbleed: Mitigation; Enhanced IBRS
Spec intack overflow: Not affected
Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1: Mitigation; usercopy/swap barriers and jsr Pointer sanitization
Spectre v2: V2
Speculation control: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PMSB=IBRS SW sequence; BMI1,BMI2,S
Taa: Not affected
Tsc syncv abort: Not affected
Vsyscall: Not affected

```

Explanation:

The `lscpu` command displays detailed information about the processor. The output shows that the system uses a 13th Gen Intel Core i7-13700HX processor with 24 logical CPUs. It also provides information about CPU architecture, cache memory, virtualization support, and processor features.

Command: ps

Output:

```
karthik@Karthikeya:~$ ps
  PID TTY          TIME CMD
  386 pts/0        00:00:00 bash
 1336 pts/0        00:00:00 ps
```

Explanation:

The `ps` command displays the processes currently running in the terminal session. In the output, the bash shell and the `ps` command itself are listed along with their Process IDs (PIDs).

Command: top

Output:

